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HAMILTONICITY IN GRAPHS DEFINED BY PRIMES AND PRIMITIVE ELEMENTS

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Abstract. A prime circle of order $2n$ is a circular ordering of $1, \dots, 2n$ such that the sum of every two adjacent terms is prime. We prove that a prime circle exists for every sufficiently large n. The proof is based on a perfect matching and robust expansion. We also study Hamilton cycles in graphs and digraphs defined by primitive sums and differences over finite fields. In particular, the primitive-sum graph on $\mathbb{F}_q$ is Hamiltonian for every prime power $q > 18\,888\,871$, and for the graph on a full prime field $\mathbb{F}_p$, the bound improves to $p > 61$.

1. Introduction

Questions concerning permutations satisfying arithmetic restrictions have a long history in combinatorics and number theory. For a positive integer m, write

$$[m] = \{1, 2, \dots, m\}.$$

A prime circle of order $2n$ is a circular ordering of $[2n]$ such that the sum of every two adjacent terms is prime. In 1982, Filz [4] posed the prime circle conjecture, which asserts that a prime circle of order $2n$ exists for every positive integer n. The conjecture is also recorded in Guy's book [7].

As usual, we denote by $V(G)$ and $E(G)$ the vertex set and the edge set of a graph G respectively. For each positive integer n, let

$$O_n = \{1, 3, \dots, 2n-1\}, \qquad E_n = \{2, 4, \dots, 2n\},$$

and define the prime-sum graph Γ_n on $[2n]$ by

$$xy \in E(\Gamma_n) \quad \Longleftrightarrow \quad x \neq y \text{ and } x + y \text{ is prime.}$$

Apart from the sum $1 + 1 = 2$, parity makes Γ_n bipartite with parts O_n and E_n. For $n \geq 2$, a prime circle of order $2n$ is precisely a Hamilton cycle in Γ_n.

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Greenfield and Greenfield [6] proved that the prime-sum graph on $[m]$ has a perfect matching if and only if m is even. Chen, Fu and Guo [1] proved that a prime circle of order $2n$ exists if there are primes $p_1 < p_2 \le 2n$ such that $2n + p_1$ and $2n + p_2$ are prime and

$$\gcd\Big(\frac{p_2 - p_1}{2}, n\Big) = 1.$$

Together with bounded gaps between primes, this gives prime circles for infinitely many orders. More recently, Shang, Li and Zhang [10] constructed a two-sided ordering of all positive integers in which the sum of every two adjacent terms is prime. The question whether a prime circle of order $2n$ exists for every positive integer n remains open. Our first result proves the conjecture for all sufficiently large n.

Theorem 1.1. *There is an integer n_0 such that Γ_n is Hamiltonian for every $n \ge n_0$. Equivalently, a prime circle of order $2n$ exists for every sufficiently large n.*

Let q be a prime power and $\mathbb{F}_q$ be the finite field of q elements. Write $\mathcal{P}_q$ for the set of primitive elements of $\mathbb{F}_q$, that is, the generators of the multiplicative group $\mathbb{F}_q^* = \mathbb{F}_q \setminus \{0\}$. Define the graph H_q on $\mathbb{F}_q$ by

$$xy \in E(H_q) \quad \Longleftrightarrow \quad x \ne y \ \text{ and } \ x + y \in \mathcal{P}_q.$$

For an odd prime $p = 2m + 1$, define a graph H_p^+ and a digraph H_p^-, both on $[m]$, by

$$xy \in E(H_p^+) \quad \Longleftrightarrow \quad x \ne y \ \text{ and } \ x + y \in \mathcal{P}_p,$$

and

$$xy \in E(H_p^-) \quad \Longleftrightarrow \quad x - y \in \mathcal{P}_p.$$

Sun and Hou [11] conjectured that these three graphs are Hamiltonian, apart from a finite list of small fields. Our second result is as follows.

Theorem 1.2. *The following statements hold.*

1. *If $q > 18\,888\,871$ is a prime power, then H_q is Hamiltonian.*
2. *If $p > 562\,582\,021$ is prime, then H_p^+ is Hamiltonian.*
3. *If $p > 3.661 \times 10^{52}$ is prime, then H_p^- contains a directed Hamilton cycle.*

Remark 1.3. For H_p, a direct construction gives a stronger result. In fact, by Cohen, Oliveira e Silva and Trudgian [3, Theorem 1], every prime $p > 61$ has an element $x \in \mathbb{F}_p$ for which $x, x+1, x+2$ are primitive. Put $m = (p-1)/2$ and define

$$a_{2k+1} = \frac{x}{2} + 2k + 1 \quad (0 \le k \le m),$$

and

$$a_{2k} = \frac{x}{2} - 2k + 1 \quad (1 \le k \le m).$$

One may easily verify that this ordering is a Hamilton cycle of H_p.

2. Graph-theoretic and analytic tools for the prime-circle problem

For a digraph D on N vertices, a set $S \subseteq V(D)$ and a real number $\nu > 0$, define the robust outneighborhood

$$\mathrm{RN}^+_{\nu,D}(S) = \{v \in V(D) : |N^-_D(v) \cap S| \ge \nu N\},$$

where $N^-_D(v)$ denotes the inneighborhood of v. The digraph is a *robust* (ν, τ)-*outexpander*, where $0 < \nu \le \tau < 1$ if

$$|\mathrm{RN}^+_{\nu,D}(S)| > |S| + \nu N$$

whenever $\tau N \le |S| \le (1-\tau)N$. The minimum semidegree of D is

$$\delta^0(D) = \min\{\delta^+(D), \delta^-(D)\},$$

where $\delta^+(D)$ and $\delta^-(D)$ are respectively the minimum outdegree and minimum indegree of D.

We use the following theorem of Lo and Patel [9, Theorem 3].

Theorem 2.1. *Let N be a positive integer and let $\nu, \tau, \eta \in (0,1)$ satisfy*

$$4\left(\frac{\log^2 N}{N}\right)^{1/13} < \nu \le \tau \le \frac{\eta}{16} < \frac{1}{16}.$$

Let D be an N-vertex digraph with $\delta^0(D) > \eta N$ which is a robust (ν, τ)-outexpander. Then, for every $\nu N/2 \le \ell \le N$ and every vertex v, the digraph D contains a directed cycle of length ℓ through v.

We next record the three number-theoretic lemmas used for prime circles. The first gives a uniform estimate for the vertex degrees of Γ_n.

Lemma 2.2. *Uniformly for $v \in V(\Gamma_n)$,*

$$d_{\Gamma_n}(v) = (2 + o(1))\frac{n}{\log n}.$$

Proof. As usual, $\pi(x)$ counts the number of primes not exceeding x. The prime number theorem gives

$$\pi(x) = \frac{x}{\log x} + O\left(\frac{x}{\log^2 x}\right).$$

For each $v \in V(\Gamma_n)$, the neighbors of v correspond to the primes in $(v, v+2n] \setminus \{2v\}$. Thus

$$d_{\Gamma_n}(v) = \pi(v+2n) - \pi(v) + O(1).$$

Put $A = n/(\log n)^2$. If $v < A$, then

$$\pi(v) \le v = o\left(\frac{n}{\log n}\right),$$

while

$$\pi(v+2n) = (1+o(1))\frac{v+2n}{\log(v+2n)} = (2+o(1))\frac{n}{\log n}.$$

If $v \ge A$, note that

$$\log t = (1+o(1))\log n \qquad (A \le t \le 4n).$$

Hence we obtain

$$\pi(v+2n)-\pi(v) = (1+o(1))\frac{v+2n}{\log n} - (1+o(1))\frac{v}{\log n} = (2+o(1))\frac{n}{\log n}.$$

Thus the estimate holds uniformly for $1 \le v \le 2n$. □

We use the upper bound sieve for prime tuples; see, for example, [8, Theorem 5.7].

Theorem 2.3. *Let g be a positive integer, and let $a_i, b_i (i = 1, \dots, g)$ be integers satisfying*

$$E := \prod_{i=1}^{g} a_i \prod_{1 \le r < s \le g} (a_r b_s - a_s b_r) \ne 0.$$

Let $\rho(p)$ denote the number of solutions of

$$\prod_{i=1}^{g}(a_i k + b_i) \equiv 0 \pmod{p},$$

and suppose that

$$\rho(p) < p \text{ for all } p.$$

Then the number of integers k in an interval of length y for which all the $a_i k + b_i$ are prime is at most

$$2^g g! \prod_p \left(1 - \frac{\rho(p)}{p}\right)\left(1 - \frac{1}{p}\right)^{-g} \frac{y}{\log^g y} \times \left(1 + O_g\left(\frac{\log\log 3y + \log\log 3|E|}{\log y}\right)\right).$$

Lemma 2.4. *If $x \ne x'$ lie in the same part of Γ_n, then*

$$\operatorname{codeg}_{\Gamma_n}(x, x') \ll \frac{n}{\log^2 n} \prod_{\substack{p \mid x - x' \\ p > 2}} \frac{p-1}{p-2} \ll \frac{n(\log\log n)^2}{\log^2 n}.$$

Proof. Take $g = 2$, $1 \le k \le n$, $a_1 = a_2 = 2$, $b_1 = x + \delta$ and $b_2 = x' + \delta$, where $\delta = 0$ or -1 according as $x \in O_n$ or $x \in E_n$. Then

$$E = 8(x' - x) \neq 0.$$

Moreover,

$$\rho(2) = 0, \qquad \rho(p) = \begin{cases} 1, & p \mid x' - x, \\ 2, & p \nmid x' - x \end{cases} \qquad (p > 2).$$

Consequently, Theorem 2.3 gives the first estimate. Since $|x - x'| \le 2n$, the second follows from

$$\prod_{\substack{p \mid x - x' \\ p > 2}} \frac{p-1}{p-2} \le \left(\frac{|x - x'|}{\varphi(|x - x'|)} \right)^2 \ll (\log \log n)^2,$$

where φ denotes the Euler totient function. □

The von Mangoldt function $\Lambda(m)$ is defined for positive integers m by

$$\Lambda(m) = \begin{cases} \log p, & \text{if } m = p^k \text{ for some prime } p \text{ and integer } k \ge 1, \\ 0, & \text{otherwise.} \end{cases}$$

For $w \ge 2$, let

$$P(w) = \prod_{p < w} p, \qquad \Lambda_{\mathrm{Cr},w}(m) = \begin{cases} \frac{P(w)}{\varphi(P(w))} & \text{if } (m, P(w)) = 1, \\ 0 & \text{otherwise.} \end{cases}$$

For a function $f \colon \mathbb{Z} \to \mathbb{C}$ and a positive integer N, the local Gowers uniformity norm $\|f\|_{U^2[N]}$ is defined by

$$\left(\mathbb{E}_{\substack{n, h_1, h_2 \in \mathbb{Z} \\ n, n+h_1, n+h_2, n+h_1+h_2 \in [N]}} f(n) \overline{f(n+h_1) f(n+h_2)} f(n+h_1+h_2) \right)^{1/4},$$

where $\mathbb{E}$ denotes the average over all triples satisfying the displayed conditions.

The following is the von Mangoldt-function part of the uniformity estimate of Tao and Teräväinen [12, Theorem 1.3(ii)].

Theorem 2.5. *There is an absolute constant $c_0 > 0$ such that, for all sufficiently large N and every*

$$2 \le w \le \exp\Big((\log N)^{1/10} \Big),$$

we have

$$\|\Lambda - \Lambda_{\mathrm{Cr},w}\|_{U^2[N]} \ll_{\mathrm{ineff}} (\log N)^{-c_0} + w^{-c_0}.$$

The subscript ineff *means the implied constants are permitted to be ineffective.*

Lemma 2.6. *There is an absolute constant $c > 0$ such that the following holds for all sufficiently large N. Let $X, Y \subseteq [N]$ and suppose that $x + y \le N$ for every $(x, y) \in X \times Y$. With*

$$w = (\log N)^{1/100},$$

we have

$$\sum_{x\in X}\sum_{y\in Y}\Lambda(x+y) = \sum_{x\in X}\sum_{y\in Y}\Lambda_{\mathrm{Cr},w}(x+y) + O_{\mathrm{ineff}}\Big(N(\log N)^{-c}\sqrt{|X||Y|}\Big).$$

Proof. Put

$$f = (\Lambda - \Lambda_{\mathrm{Cr},w})\mathbf{1}_{[N]}.$$

We write $\mathrm{e}(t) = e^{2\pi i t}$ and define

$$\widehat{f}(\theta) = \sum_{m\in\mathbb{Z}} f(m)\mathrm{e}(m\theta).$$

Cauchy–Schwarz's inequality gives

$$\begin{aligned}|\widehat{f}(\theta)|^4 &= |\sum_{-N+1\le h\le N-1} e(h\theta)\sum_{m\in\mathbb{Z}} f(m+h)\overline{f(m)}|^2\\ &\le (2N-1)\sum_{h\in\mathbb{Z}}|\sum_{m\in\mathbb{Z}} f(m+h)\overline{f(m)}|^2\\ &\ll N^4\|\Lambda - \Lambda_{\mathrm{Cr},w}\|_{U^2[N]}^4.\end{aligned}$$

Consequently, by Theorem 2.5,

$$\sup|\widehat{f}(\theta)| \ll N\|\Lambda - \Lambda_{\mathrm{Cr},w}\|_{U^2[N]} \ll_{\mathrm{ineff}} N\Big((\log N)^{-c_0} + w^{-c_0}\Big)$$

for some constant c_0. For $w = (\log N)^{1/100}$, this becomes

$$\sup|\widehat{f}(\theta)| \ll_{\mathrm{ineff}} N(\log N)^{-c}, \qquad c = \frac{c_0}{100}.$$

Furthermore, Hölder's inequality gives

$$\begin{aligned}&\left|\sum_{x\in X}\sum_{y\in Y} f(x+y)\right|\\ &= \left|\int_0^1 \widehat{f}(\theta)\left(\sum_{x\in X}\mathrm{e}(-x\theta)\right)\left(\sum_{y\in Y}\mathrm{e}(-y\theta)\right)\,\mathrm{d}\theta\right|\\ &\le \sup|\widehat{f}(\theta)|\left(\int_0^1|\sum_{x\in X}\mathrm{e}(-x\theta)|^2 d\theta\right)^{1/2}\left(\int_0^1|\sum_{y\in Y}\mathrm{e}(-y\theta)|^2 d\theta\right)^{1/2}\\ &\ll_{\mathrm{ineff}} N(\log N)^{-c}\sqrt{|X||Y|}.\end{aligned}$$

The result follows from the definition of f. □

3. Edge-count estimates for the prime-sum graph

Recall that $N_G(v)$ denotes the neighborhood of a vertex v in the graph G. For a graph or digraph G and arbitrary sets $X, Y \subset V(G)$, define

$$e_G(X,Y) = \Big|\{(x,y) \in X \times Y : xy \in E(G)\}\Big|.$$

Choose a sufficiently small absolute constant $\rho_0 > 0$ and put

$$\rho_n = \frac{\rho_0}{(\log\log n)^2}.$$

Lemma 3.1. *For all sufficiently large n, if X lies in one part of Γ_n and U lies in the opposite part with $|X| \le \rho_n n$ and $|U| \le 2|X|$, then*

$$e_{\Gamma_n}(X,U) \le \frac{n}{2\log n}|X|.$$

Proof. The assertion is trivial when $x = \emptyset$, so assume $X \neq \emptyset$. Given a vertex y, write

$$d_X(y) = |N_{\Gamma_n}(y) \cap X|.$$

Let δ_{xy} be 1 or 0 according as $xy \in E(\Gamma_n)$ or not. Note that

$$\sum_y d_X(y)^2 = \sum_y \left(\sum_{x\in X} \delta_{xy}\right)^2 = \sum_{x\in X} d_{\Gamma_n}(x) + \sum_{\substack{x,x'\in X\\ x\neq x'}} \mathrm{codeg}_{\Gamma_n}(x,x').$$

Hence by Lemma 2.2 and Lemma 2.4, there is a constant C such that

$$\sum_y d_X(y)^2 \le C|X|\left(\frac{n}{\log n} + \frac{n(\log\log n)^2|X|}{\log^2 n}\right).$$

By Cauchy–Schwarz's inequality, we have

$$\begin{aligned} e_{\Gamma_n}(X,U)^2 &\le |U| \sum_y d_X(y)^2 \\ &\le 2C|X|^2\left(\frac{n}{\log n} + \frac{n(\log\log n)^2|X|}{\log^2 n}\right). \end{aligned}$$

After division by $(n|X|/\log n)^2$, by the upper bound for $|X|$, the right-hand side is at most

$$2C\left(\frac{\log n}{n} + \rho_0\right).$$

Choose ρ_0 sufficiently small in terms of C and then take n sufficiently large. This proves the assertion. □

Lemma 3.2. *Let Q be an odd square-free integer divisible by 3, and let ε_Q be the principal Dirichlet character modulo Q. Let K_Q be the $Q \times Q$ matrix with (i,j)-entry $\varepsilon_Q(i+j)$. Then its largest singular value*

is $\varphi(Q)$, with constant singular vectors, and every other singular value is at most $\varphi(Q)/2$.

Proof. For $1 \le r \le Q$, define the vectors

$$v_r = \frac{1}{\sqrt{Q}} \left(\mathrm{e}\left(\frac{r}{Q}\right), \mathrm{e}\left(\frac{2r}{Q}\right), \dots, \mathrm{e}\left(\frac{rQ}{Q}\right) \right)^{\mathsf{T}}.$$

Note that for any $1 \le r, j \le Q$

$$\sum_{k=1}^{Q} \varepsilon_Q(j+k) \mathrm{e}\left(\frac{rk}{Q}\right) = c_Q(r) \mathrm{e}\left(\frac{-rj}{Q}\right),$$

where $c_Q(r)$ are the Ramanujan sums

$$\sum_{\substack{u \bmod Q \\ (u,Q)=1}} \mathrm{e}\left(\frac{ru}{Q}\right).$$

Hence

$$K_Q v_r = c_Q(r) \overline{v_r}.$$

Since the matrix with column vectors $v_1, v_2, \dots, v_Q$ is unitary, the singular values of K_Q are the absolute values of $c_Q(r)$. Since Q is square-free,

$$|c_Q(r)| = \frac{\varphi(Q)}{\varphi(Q/(Q,r))}.$$

For $r = Q$ this is $\varphi(Q)$. If $r \neq Q$, then $Q/(Q,r) > 1$. This divisor is odd, so its totient is at least 2. The assertion follows. □

Lemma 3.3. *Let*

$$w = (\log 4n)^{1/100}, \qquad Q = \frac{P(w)}{2}.$$

Suppose $X \subseteq O_n$ and $Y \subseteq E_n$ with

$$|X|, |Y| \ge \rho_n n, \qquad |X| + |Y| \ge n - \frac{n}{\log^2 n}.$$

Then, for all sufficiently large n,

$$\sum_{\substack{a,b \bmod Q \\ (a+b,Q)=1}} |X \cap (a \bmod Q)||Y \cap (b \bmod Q)| \ge \frac{\varphi(Q)}{3Q} |X||Y|.$$

Proof. Let A and B be Q-dimensional vectors with entries

$$A_i = |X \cap (i \bmod Q)|, \quad B_i = |Y \cap (i \bmod Q)|, \quad 1 \le i \le Q.$$

It is trivial that

$$0 \le A_i, B_i \le \frac{n}{Q} + 1.$$

Let $\mathbf{1}$ denote the all-ones vector. Write

$$A = \frac{|X|}{Q}\mathbf{1} + A_0, \qquad B = \frac{|Y|}{Q}\mathbf{1} + B_0.$$

It is easy to see that A_0, B_0 are orthogonal to the constant vector. By Lemma 3.2,

$$A^{\mathsf{T}} K_Q B \geq \frac{\varphi(Q)}{Q}|X||Y| - \frac{\varphi(Q)}{2}\|A_0\|_2\|B_0\|_2.$$

Write

$$|X| = \alpha n, \qquad |Y| = \beta n.$$

The above bound for A_i gives

$$\begin{aligned}\|A_0\|_2^2 &= \sum_{i=1}^{Q} A_i^2 - \frac{|X|^2}{Q} \\ &\leq \left(\frac{n}{Q} + 1\right)|X| - \frac{|X|^2}{Q} \\ &= \frac{n^2}{Q}\alpha\left(1 - \alpha + \frac{Q}{n}\right),\end{aligned}$$

and similarly for B_0.

Put $\delta = Q/n$. The elementary estimate $\log Q = O(w)$ gives $\delta = n^{-1+o(1)}$. Also, for large n, $\alpha\beta \geq \max\{\alpha, \beta\}\rho_n \geq \rho_n/3$. Finally,

$$\begin{aligned}A^{\mathsf{T}} K_Q B &\geq \frac{\varphi(Q)}{Q}|X||Y|\left(1 - \frac{1}{2}\sqrt{\frac{(1-\alpha+\delta)(1-\beta+\delta)}{\alpha\beta}}\right) \\ &\geq \frac{\varphi(Q)}{Q}|X||Y|\left(1 - \frac{1}{2}\sqrt{1 + (\alpha\beta)^{-1}\left(\frac{1}{\log^2 n} + 2\delta + \delta^2\right)}\right) \\ &= \frac{\varphi(Q)}{Q}|X||Y|\left(\frac{1}{2} - o(1)\right),\end{aligned}$$

which is stronger than the stated bound. □

Lemma 3.4. *If $X \subseteq O_n$ and $Y \subseteq E_n$ satisfy*

$$|X|, |Y| \geq \rho_n n, \qquad |X| + |Y| \geq n - \frac{n}{\log^2 n},$$

then there is an absolute constant $c_1 > 0$ such that for all sufficiently large n,

$$e_{\Gamma_n}(X, Y) \geq c_1 \frac{|X||Y|}{\log n}.$$

Proof. Take $N = 4n$ and $w = (\log N)^{1/100}$, and write $P(w) = 2Q$ as in Lemma 3.3. Since every $x + y$ with $(x, y) \in O_n \times E_n$ is odd,

$$(x + y, P(w)) = 1 \quad \Longleftrightarrow \quad (x + y, Q) = 1.$$

Thus Lemma 3.3 yields

$$\sum_{x\in X}\sum_{y\in Y} \Lambda_{\mathrm{Cr},w}(x + y) \geq \frac{2}{3}|X||Y|.$$

By Lemma 2.6, the error in replacing $\Lambda_{\mathrm{Cr},w}$ by Λ is

$$O_{\mathrm{ineff}}\Big(n(\log n)^{-c}\sqrt{|X||Y|}\Big) = o(|X||Y|).$$

Indeed, the hypotheses give $|X||Y| \geq (\rho_n/3)n^2$.

It remains to remove prime powers. Put

$$\vartheta(m) = \begin{cases} \log m, & m \text{ is prime,} \\ 0, & \text{otherwise.} \end{cases}$$

If

$$r_{X,Y}(m) = |\{(x, y) \in X \times Y : x + y = m\}|,$$

then $r_{X,Y}(m) \leq n$, and hence

$$\begin{aligned}\sum_{x\in X}\sum_{y\in Y}\Big(\Lambda - \vartheta\Big)(x + y) &\leq n \sum_{\substack{p^k\leq 4n\\ k\geq 2}} \log p \\ &\ll n^{3/2}\log n = o(|X||Y|).\end{aligned}$$

Therefore, for all sufficiently large n,

$$\sum_{x\in X}\sum_{y\in Y} \vartheta(x + y) \geq \frac{1}{3}|X||Y|.$$

Since every relevant prime is at most $4n$, division by $\log 4n$ proves the result. □

4. Proof of Theorem 1.1

The following matching theorem is due to Greenfield and Greenfield [6, Theorem 1].

Theorem 4.1. *For every $n \geq 1$, the graph Γ_n has a perfect matching.*

Fix a perfect matching

$$M = \{o_i e_i : 1 \leq i \leq n\}, \qquad o_i \in O_n, \quad e_i \in E_n.$$

Define a loopless digraph D_M on $[n]$ by

$$ij \in E(D_M) \quad \Longleftrightarrow \quad i \neq j \ \text{ and } \ o_i + e_j \text{ is prime.}$$

Lemma 4.2. *If D_M has a directed Hamilton cycle, then Γ_n has a Hamilton cycle.*

Proof. If $i_1 \to i_2 \to \cdots \to i_n \to i_1$ is a directed Hamilton cycle in D_M, then

$$e_{i_1}, o_{i_1}, e_{i_2}, o_{i_2}, \ldots, e_{i_n}, o_{i_n}, e_{i_1}$$

is a Hamilton cycle in Γ_n. □

Lemma 4.3. *For all sufficiently large n and every choice of M,*

$$\delta^0(D_M) \geq \frac{n}{\log n}.$$

Proof. For each i,

$$d^+_{D_M}(i) = d_{\Gamma_n}(o_i) - 1, \qquad d^-_{D_M}(i) = d_{\Gamma_n}(e_i) - 1.$$

The deleted edge is exactly $o_i e_i$. The result follows from Lemma 2.2. □

Fix a sufficiently small absolute constant $\kappa > 0$ and put

$$\eta = \frac{1}{2\log n}, \qquad \tau = \frac{1}{32\log n}, \qquad \nu = \frac{\kappa}{\log^2 n}. \tag{4.1}$$

Lemma 4.4. *For all sufficiently large n and every perfect matching M, the digraph D_M is a robust (ν, τ)-outexpander.*

Proof. Let $S \subseteq [n]$ satisfy

$$\tau n \leq s := |S| \leq (1-\tau)n.$$

Suppose for a contradiction that, with

$$R = \mathrm{RN}^+_{\nu, D_M}(S), \qquad T = [n] \setminus R, \qquad t = |T|,$$

we have $|R| \leq s + \nu n$. Then

$$s + t \geq n - \nu n, \qquad t \geq (\tau - \nu)n \geq \frac{\tau n}{2}$$

for large n. The definition of T gives

$$e_{D_M}(S, T) < \nu n t.$$

Let

$$X = \{o_i : i \in S\} \subseteq O_n, \qquad Y = \{e_j : j \in T\} \subseteq E_n.$$

Only matching edges with an index in $S \cap T$ are counted in $e_{\Gamma_n}(X, Y)$ but not in $e_{D_M}(S, T)$. Consequently,

$$e_{\Gamma_n}(X, Y) < \nu n t + n. \tag{4.2}$$

We distinguish three cases.

Case 1: $s \leq \rho_n n$. The set $U = E_n \setminus Y$ satisfies

$$|U| = |R| \leq s + \nu n \leq 2s$$

for large n, since $s \geq \tau n$ and $\nu = o(\tau)$. By Lemma 2.2 and Lemma 3.1,

$$e_{\Gamma_n}(X, Y) \geq \frac{1}{2}\frac{sn}{\log n} \geq \frac{n^2}{64 \log^2 n}.$$

This contradicts (4.2) when, for example, $\kappa < 1/128$ and n is large.

Case 2: $t \leq \rho_n n$. Now $U = O_n \setminus X$ has

$$|U| = n - s \leq t + \nu n \leq 2t.$$

Applying Lemma 2.2 and Lemma 3.1 with Y in place of X gives

$$e_{\Gamma_n}(X, Y) \geq \frac{1}{2}\frac{tn}{\log n}.$$

This again contradicts (4.2), because

$$\frac{\nu n t + n}{tn/\log n} \leq \frac{\kappa}{\log n} + O\left(\frac{\log^2 n}{n}\right) = o(1).$$

Case 3: $s, t > \rho_n n$. The hypotheses of Lemma 3.4 hold, so

$$e_{\Gamma_n}(X, Y) \geq c_1 \frac{st}{\log n}.$$

This is incompatible with (4.2), since

$$\frac{c_1 st/\log n}{\nu n t} = \frac{c_1 (s/n) \log n}{\kappa} \geq \frac{c_1 \rho_0 \log n}{\kappa (\log \log n)^2} \longrightarrow \infty,$$

and the additive term n is negligible. Every case is contradictory, which proves robust outexpansion. □

Proof of Theorem 1.1. Choose a perfect matching M by Theorem 4.1 and form D_M. By Lemma 4.3, we have

$$\delta^0(D_M) > \frac{n}{2 \log n} = \eta n.$$

By Lemma 4.4, D_M is a robust (ν, τ)-outexpander. The parameters in (4.1) satisfy

$$\nu \leq \tau = \frac{\eta}{16} < \frac{1}{16}$$

and

$$4\left(\frac{\log^2 n}{n}\right)^{1/13} < \frac{\kappa}{\log^2 n} = \nu$$

for all sufficiently large n. Hence Theorem 2.1 gives a directed Hamilton cycle in D_M. By Lemma 4.2, this cycle lifts to a Hamilton cycle of Γ_n. □

5. Preliminaries for Theorem 1.2

For a positive integer r, let $\omega(r)$ be the number of distinct prime divisors of r, and put

$$W(r) = 2^{\omega(r)}.$$

Thus $W(r)$ is the number of square-free divisors of r. We write μ for the Möbius function. Also, we define the radical of r by

$$\mathrm{Rad}(r) = \prod_{\substack{\ell \mid r \\ \ell \text{ prime}}} \ell.$$

Let $e \mid q-1$. An element $x \in \mathbb{F}_q$ is *e-free* if $x \neq 0$ and $x = y^d$ with $d \mid e$ implies $d = 1$. In particular, x is primitive exactly when it is $(q-1)$-free. Let $\widehat{\mathbb{F}_q^*}$ and $\widehat{\mathbb{F}_q}$ denote the groups of multiplicative and additive characters of $\mathbb{F}_q$, respectively. We extend every multiplicative character of $\mathbb{F}_q^*$ to 0 by setting $\chi(0) = 0$. The standard characteristic-function is

$$\mathbf{1}_{e\text{-free}}(x) = \frac{\varphi(e)}{e} \sum_{d \mid e} \frac{\mu(d)}{\varphi(d)} \sum_{\substack{\chi \in \widehat{\mathbb{F}_q^*} \\ \mathrm{ord}\,\chi = d}} \chi(x). \tag{5.1}$$

Lemma 5.1. *Let $e \mid q-1$ and let ψ be a nontrivial additive character of $\mathbb{F}_q$. Then*

$$\left| \sum_{x \ e\text{-free}} \psi(x) \right| \leq \frac{\varphi(e)}{e} W(e) \sqrt{q}.$$

Proof. Substituting (5.1), we obtain

$$\sum_{\substack{x \in \mathbb{F}_q \\ x \ e\text{-free}}} \psi(x) = \frac{\varphi(e)}{e} \sum_{d \mid e} \frac{\mu(d)}{\varphi(d)} \sum_{\substack{\chi \in \widehat{\mathbb{F}_q^*} \\ \mathrm{ord}(\chi) = d}} \sum_{x \in \mathbb{F}_q} \chi(x)\psi(x).$$

For the principal multiplicative character, the inner sum equals -1. For every nonprincipal multiplicative character it is a Gauss sum of absolute value $\sqrt{q}$. There are $\varphi(d)$ characters of order d, and only square-free divisors d contribute. Taking absolute values gives the bound. □

Put

$$\Lambda_q = \frac{\varphi(q-1)}{q-1} W(q-1) \sqrt{q}.$$

For $X, Y \subseteq \mathbb{F}_q$, put

$$E^+(X,Y) = |\{(x,y) \in X \times Y : x + y \in \mathcal{P}_q\}|,$$
$$E^-(X,Y) = |\{(x,y) \in X \times Y : x - y \in \mathcal{P}_q\}|.$$

Lemma 5.2. *For all $X, Y \subseteq \mathbb{F}_q$,*

$$\left| E^+(X,Y) - \frac{\varphi(q-1)}{q}|X||Y| \right| \leq \Lambda_q \sqrt{|X||Y|},$$

and

$$\left| E^-(X,Y) - \frac{\varphi(q-1)}{q}|X||Y| \right| \leq \Lambda_q \sqrt{|X||Y|}.$$

Proof. Recall that for a function $f : \mathbb{F}_q \to \mathbb{C}$, Fourier inversion gives

$$f(t) = \frac{1}{q} \sum_{\psi \in \widehat{\mathbb{F}_q}} \widehat{f}(\psi)\psi(t),$$

where

$$\widehat{f}(\psi) = \sum_{u \in \mathbb{F}_q} f(u)\overline{\psi(u)}.$$

Let $f = \mathbf{1}_{(q-1)\text{-free}}$. By the definition of $E^+(X,Y)$ and Fourier inversion, we have

$$\begin{aligned} E^+(X,Y) &= \sum_{x \in X} \sum_{y \in Y} f(x+y) \\ &= \frac{1}{q} \sum_{\psi \in \widehat{\mathbb{F}_q}} \widehat{f}(\psi) \left(\sum_{x \in X} \psi(x) \right) \left(\sum_{y \in Y} \psi(y) \right). \end{aligned}$$

The trivial additive character contributes

$$\frac{\varphi(q-1)|X||Y|}{q}.$$

If ψ is nontrivial, Lemma 5.1 gives $|\widehat{f}(\psi)| \leq \Lambda_q$. Cauchy's inequality then yields

$$\begin{aligned} \left| E^+(X,Y) - \frac{\varphi(q-1)|X||Y|}{q} \right| &\leq \frac{\Lambda_q}{q} \sum_{\substack{\psi \in \widehat{\mathbb{F}_q} \\ \psi \text{ nontrivial}}} \left| \sum_{x \in X} \psi(x) \right| \left| \sum_{y \in Y} \psi(y) \right| \\ &\leq \frac{\Lambda_q}{q} \sqrt{(q|X| - |X|^2)(q|Y| - |Y|^2)} \\ &\leq \Lambda_q \sqrt{|X||Y|}. \end{aligned}$$

For $E^-(X,Y)$ one uses $\psi(x-y) = \psi(x)\overline{\psi(y)}$, and the same argument applies. □

An interval in $\mathbb{F}_p$ is a set of consecutive residues. Put

$$L(x) = \frac{2}{\pi^2} \log x + 1.$$

The explicit Pólya–Vinogradov inequality of Frolenkov and Soundararajan [5, Theorem 2] implies that, for $p > 1200$, every nonprincipal character χ modulo p and every interval $I \subseteq \mathbb{F}_p$ satisfy

$$\left|\sum_{a\in I}\chi(a)\right| \le \sqrt{p}\,L(p). \tag{5.2}$$

Lemma 5.3. *Let $p > 1200$ be prime, let $I \subseteq \mathbb{F}_p$ be an interval, and put $I^* = I \setminus \{0\}$. Then*

$$\left||\mathcal{P}_p \cap I| - \frac{\varphi(p-1)}{p-1}|I^*|\right| \le \frac{\varphi(p-1)}{p-1}W(p-1)\sqrt{p}\,L(p).$$

Proof. Sum (5.1) over I with $e = p-1$. The principal character contributes $\varphi(p-1)|I^*|/(p-1)$. Bound every nonprincipal contribution by (5.2) and the stated estimate follows. □

For undirected graphs we use the following theorem of Chvátal and Erdős [2, Theorem 1].

Theorem 5.4. *Let G be a finite simple graph on at least three vertices. If its vertex connectivity $\kappa(G)$ is at least its independence number $\alpha(G)$, then G is Hamiltonian.*

6. Primitive-sum graphs on a finite field

We first prove the finite-field part of Theorem 1.2. The argument is a direct comparison between independence and connectivity.

Proof of Theorem 1.2(1). Let I be an independent set of size s in H_q. Partition I into disjoint sets X, Y with sizes $\lfloor s/2\rfloor$ and $\lceil s/2\rceil$. Since $E^+(X,Y) = 0$, Lemma 5.2 gives

$$\sqrt{|X||Y|} \le \frac{\Lambda_q q}{\varphi(q-1)}.$$

It follows that

$$\alpha(H_q) \le \frac{2\Lambda_q q}{\varphi(q-1)} + 1. \tag{6.1}$$

Suppose first that H_q is disconnected. Let A be a smallest component and put $B = \mathbb{F}_q \setminus A$. Then $|B| \ge |A|$ and the no-edge estimate as above gives

$$|A| \le \frac{\Lambda_q q}{\varphi(q-1)}.$$

This contradicts the fact that every vertex of A has degree at least $\varphi(q-1)-1$ whenever

$$\varphi(q-1)-1 > \frac{\Lambda_q q}{\varphi(q-1)}.$$

Thus the numerical condition below also ensures that H_q is connected.

Now let S be a vertex cut, let A be a smallest component of $H_q - S$, and put $B = \mathbb{F}_q \setminus (A \cup S)$. There are no edges from A to B, and $|B| \geq |A|$, so again $|A| \leq \Lambda_q q/\varphi(q-1)$. For $a \in A$, at most $|A|-1$ of its neighbors lie in A. Hence

$$\kappa(H_q) \geq (\varphi(q-1)-1) - (|A|-1) \geq \varphi(q-1) - \frac{\Lambda_q q}{\varphi(q-1)}. \tag{6.2}$$

By Theorem 5.4, equations (6.1)–(6.2) prove Hamiltonicity whenever

$$\varphi(q-1) \geq \frac{3q}{q-1} W(q-1)\sqrt{q} + 1.$$

By Lemma 9.1(1), this inequality holds for every prime power $q > 18\,888\,871$. □

7. Primitive-sum graphs on a half interval

Proof of Theorem 1.2(2). Let $p > 1200$ be prime and put $m = (p-1)/2$. The independent-set argument from Section 6, applied inside $[m]$, gives

$$\alpha(H_p^+) \leq \frac{2\Lambda_p p}{\varphi(p-1)} + 1. \tag{7.1}$$

Fix $i \in [m]$. As j ranges over $[m]$, the residues $i+j$ form an interval of length m not containing 0. By Lemma 5.3, and allowing for the forbidden loop $j = i$,

$$\delta(H_p^+) \geq \frac{\varphi(p-1)}{2} - \Lambda_p L(p) - 1.$$

The same smallest-component argument used for H_q shows, once the graph is connected, that

$$\kappa(H_p^+) \geq \delta(H_p^+) - \frac{\Lambda_p p}{\varphi(p-1)} + 1. \tag{7.2}$$

The sufficient inequality obtained by comparing (7.1) and (7.2) is

$$\varphi(p-1) \geq \frac{6W(p-1)p^{3/2} + 2p - 2}{p - 1 - 2W(p-1)\sqrt{p}\,L(p)} > 0, \tag{7.3}$$

which holds for every $p > 562\,582\,021$ by Lemma 9.1(2). Indeed, this condition gives $\delta(H_p^+) > \Lambda_p p/\varphi(p-1)$, so it also supplies the connectivity used in (7.2). The Chvátal–Erdős theorem completes the proof. □

8. Primitive-difference digraphs

Proof of Theorem 1.2(3). Let $p > 3.661 \times 10^{52}$ be prime, set $m = (p-1)/2$, and put

$$\sigma = \frac{9999}{10000}, \qquad \eta = \frac{\sigma\varphi(p-1)}{p}, \qquad \tau = \frac{\eta}{16}, \qquad \nu = \frac{\eta^2}{16}.$$

Then $0 < \nu < \tau = \eta/16 < 1/16$.

Let $S \subseteq [m]$ satisfy $\tau m \le |S| \le (1-\tau)m$, and put

$$T = [m] \setminus \mathrm{RN}^+_{\nu,H_p^-}(S).$$

Then

$$E^-(S,T) < \nu m|T|.$$

Suppose that $|\mathrm{RN}^+_{\nu,H_p^-}(S)| \le |S| + \nu m$. Then

$$|T| \ge (\tau - \nu)m.$$

On the other hand, Lemma 5.2 gives

$$E^-(S,T) \ge |T| \left(\frac{\tau m \varphi(p-1)}{p} - \Lambda_p \sqrt{\frac{1-\tau}{\tau-\nu}} \right).$$

The first inequality in Lemma 9.1(3) is precisely the condition that the parenthesis is larger than νm. This is a contradiction. Thus H_p^- is a robust (ν,τ)-outexpander.

For a fixed $i \in [m]$, each of the sets $i - [m]$ and $[m] - i$ is an interval of length m containing 0. By Lemma 5.3,

$$\delta^0(H_p^-) \ge \frac{\varphi(p-1)}{p-1} \left(\frac{p-3}{2} - W(p-1)\sqrt{p}\, L(p) \right).$$

The second inequality in Lemma 9.1(3) makes the right-hand side greater than

$$\frac{\sigma\varphi(p-1)(p-1)}{2p} = \eta m.$$

The third inequality there gives

$$\nu > 4 \left(\frac{\log^2 m}{m} \right)^{1/13}.$$

Every hypothesis of Theorem 2.1 is now satisfied, so H_p^- has a directed Hamilton cycle. □

9. Numerical verification of the finite-field thresholds

We collect here the arithmetic calculations used above.

Lemma 9.1. *Put $\sigma = 9999/10000$.*

(1) *For every prime power $q > 18\,888\,871$,*

$$\varphi(q-1) \geq \frac{3q}{q-1} W(q-1)\sqrt{q} + 1.$$

(2) *For every prime $p > 562\,582\,021$,*

$$\varphi(p-1) \geq \frac{6W(p-1)p^{3/2} + 2p - 2}{p - 1 - 2W(p-1)\sqrt{p}\,L(p)} > 0.$$

(3) *For every prime $p > 3.661 \times 10^{52}$, the following three inequalities hold:*

$$\varphi(p-1) > \frac{32p^2 W(p-1)\sqrt{p}}{\sigma(1-\sigma)(p-1)^2} \sqrt{\frac{p(16p - \sigma\varphi(p-1))}{\sigma\varphi(p-1)(p - \sigma\varphi(p-1))}}, \tag{9.1}$$

$$\frac{p-3}{2} - \frac{\sigma(p-1)^2}{2p} > W(p-1)\sqrt{p}\,L(p), \tag{9.2}$$

$$\frac{\sigma^2\varphi(p-1)^2}{16p^2} > 4\left(\frac{2\log^2((p-1)/2)}{p-1}\right)^{1/13}. \tag{9.3}$$

Proof. Let p_j be the jth prime and define

$$P_r = \prod_{j=1}^{r} p_j, \qquad \alpha_r = \frac{\varphi(P_r)}{P_r}, \qquad \beta_r = \frac{\varphi(P_r)}{\sqrt{P_r}}.$$

If $\omega(N) = r$, then

$$N \geq P_r, \qquad \frac{\varphi(N)}{N} \geq \alpha_r, \qquad \frac{\varphi(N)}{\sqrt{N}} \geq \beta_r. \tag{9.4}$$

Indeed, if $N = \prod_{j=1}^{r} \ell_j^{a_j}$ with $\ell_1 < \cdots < \ell_r$ primes and $a_1, \ldots, a_r$ positive integers, then $\ell_j \geq p_j$ and

$$\frac{\varphi(N)}{\sqrt{N}} = \prod_{j=1}^{r} \ell_j^{(a_j-1)/2} \frac{\ell_j - 1}{\sqrt{\ell_j}}.$$

Each factor is minimized when $a_j = 1$ and $\ell_j = p_j$.

Part (1). Put $N = q - 1$ and $r = \omega(N)$. After division by $\sqrt{N}$, the assertion in part (1) becomes

$$\frac{\varphi(N)}{\sqrt{N}} \geq 3 \cdot 2^r \left(1 + \frac{1}{N}\right)^{3/2} + \frac{1}{\sqrt{N}}. \tag{9.5}$$

Here $N > 18\,888\,870$. If $r \le 7$, the right-hand side is less than 385, whereas

$$\frac{\varphi(N)}{\sqrt{N}} \ge \alpha_7\sqrt{N} > \frac{3072}{17017}\sqrt{18\,888\,870} > 784.$$

Suppose $r = 8$. If N is square-free, a finite split according to the largest prime factor shows that its increasing prime tuple is coordinate-wise at least one of

$$(2,3,5,7,13,17,19,23),$$
$$(2,3,5,7,11,13,23,29), \quad (2,3,5,7,11,17,19,29),$$
$$(2,3,5,7,11,13,19,37), \quad (2,3,5,7,11,13,17,41).$$

The smallest resulting lower bound is

$$\frac{\varphi(N)}{\sqrt{N}} \ge \prod_{\ell\in\{2,3,5,7,11,13,23,29\}} \frac{\ell-1}{\sqrt{\ell}} = \frac{3\,548\,160}{\sqrt{20\,030\,010}} > 790.$$

If N is not square-free, then, apart from $N = 2P_8 = 19\,399\,380$, one has $N \ge 23\,483\,460$ and hence

$$\frac{\varphi(N)}{\sqrt{N}} \ge \alpha_8\sqrt{23\,483\,460} > 828.$$

The exceptional value would give

$$q = 19\,399\,381 = 67 \cdot 289\,543,$$

which is not a prime power. In all $r = 8$ cases, the right-hand side of (9.5) is less than 769.

Finally,

$$\beta_9 = \frac{36\,495\,360}{\sqrt{223\,092\,870}} > 4 \cdot 2^9.$$

For $r \ge 9$, each additional factor $(p_j - 1)/\sqrt{p_j}$ is greater than 2, so $\beta_r > 4 \cdot 2^r$. The right-hand side of (9.5) is smaller than $4 \cdot 2^r$. This proves part (1).

Part (2). Put

$$r = \omega(p-1), \qquad t = \frac{\varphi(p-1)}{p-1}.$$

For $u > 1$ and $a > 0$, define

$$\mathcal{H}_r(u,a) = \frac{2^{r+1}\sqrt{u}\,L(u)}{u-1} + \frac{6 \cdot 2^r u^{3/2}}{a(u-1)^2} + \frac{2}{a(u-1)}.$$

The assertion in part (2) is equivalent to

$$\mathcal{H}_r(p,t) \le 1.$$

Indeed, this inequality makes the first summand strictly smaller than 1, giving denominator positivity, and rearrangement yields the required lower bound for $\varphi(p-1)$.

Direct differentiation shows that $\mathcal{H}_r(u,a)$ decreases with $u > 1$. Clearly, it also decreases with a and increases with r. We use the following numerical evaluations, where

$$\alpha_* = \alpha_8 \frac{36}{37}.$$

Direct substitution gives

r	u	a	$\mathcal{H}_r(u,a)$
8	562 582 021	α_8	< 0.489
9	562 582 021	α_*	< 0.998
9	601 380 781	α_9	< 0.979
10	6 469 693 231	α_{10}	< 0.626

If $r \le 8$, then $t \ge \alpha_8$, and hence

$$\mathcal{H}_r(p,t) \le \mathcal{H}_8(562\,582\,021, \alpha_8) < 1.$$

Suppose next that $r = 9$, and write the distinct prime divisors of $p-1$ as $s_1 < \cdots < s_9$. Unless

$$\mathrm{Rad}(p-1) = P_8 q, \qquad q \in \{23, 29, 31\},$$

we have $t \ge \alpha_*$. Indeed, if $s_9 \ge 37$, then

$$t \ge \alpha_8 \frac{36}{37} = \alpha_*.$$

Otherwise, outside the three exceptional patterns, necessarily $s_8 \ge 23$ and $s_9 \ge 29$, so

$$t \ge \alpha_7 \frac{22}{23}\frac{28}{29} > \alpha_*.$$

Thus every nonexceptional pattern satisfies

$$\mathcal{H}_9(p,t) \le \mathcal{H}_9(562\,582\,021, \alpha_*) < 1.$$

For the three exceptional patterns, observe that

$$p - 1 > 562\,582\,020 = 2P_8 \cdot 29.$$

Since $p-1$ is a multiple of its radical, we obtain, respectively,

$$p - 1 \ge 3P_8 \cdot 23, \qquad p - 1 \ge 3P_8 \cdot 29, \qquad p - 1 \ge 2P_8 \cdot 31.$$

Consequently,

$$p \ge 2P_8 \cdot 31 + 1 = 601\,380\,781,$$

and therefore

$$\mathcal{H}_9(p,t) \le \mathcal{H}_9(601\,380\,781, \alpha_9) < 1.$$

Finally, suppose that $r \geq 10$. Then

$$\mathcal{H}_r(p,t) \leq \mathcal{H}_r(P_r+1, \alpha_r).$$

The expression on the right decreases with r. To verify this, note that Euclid's argument gives $p_{r+1} \leq P_r + 1$, whence

$$L(P_{r+1}+1) \leq 2L(P_r+1).$$

The ratios of the three summands in $\mathcal{H}_{r+1}(P_{r+1}+1, \alpha_{r+1})$ to their counterparts in $\mathcal{H}_r(P_r+1, \alpha_r)$ are at most

$$\frac{4}{\sqrt{p_{r+1}}}, \qquad \frac{2\sqrt{p_{r+1}}}{p_{r+1}-1}, \qquad \frac{1}{p_{r+1}-1},$$

respectively. All three are smaller than 1 since $p_{r+1} \geq 31$. Since $P_{10}+1 = 6\,469\,693\,231$, it follows that

$$\mathcal{H}_r(p,t) \leq \mathcal{H}_{10}(P_{10}+1, \alpha_{10}) < 0.626 < 1.$$

This proves part (2).

Part (3). Put $X = 3.661 \times 10^{52}$, $N = p-1$, and $r = \omega(N)$. We first establish (9.3). Define

$$F_r(x) = \left(\frac{\sigma\alpha_r(x-1)}{8x}\right)^{26} \frac{x-1}{2\log^2((x-1)/2)}.$$

By (9.4), it suffices to show that $F_r(p) > 1$. For $x > 2e^2+1$, the function $F_r(x)$ increases with x, while for fixed x it decreases with r. Direct evaluation gives

$$F_{32}(X) > 1.00009, \qquad F_{33}(P_{33}+1) > 1.6086.$$

Thus, if $r \leq 32$, then

$$F_r(p) \geq F_{32}(p) > F_{32}(X) > 1.$$

For $r \geq 33$, we have $p \geq P_r + 1$. We claim that $F_r(P_r+1)$ increases with r in this range. Again we have $p_{r+1} \leq P_r + 1$. Consequently,

$$\log(P_{r+1}/2) < 3\log(P_r/2).$$

Therefore, by Bernoulli's inequality, we obtain

$$\frac{F_{r+1}(P_{r+1}+1)}{F_r(P_r+1)} > \frac{p_{r+1}}{9}\left(1 - \frac{1}{p_{r+1}}\right)^{26} \geq \frac{p_{r+1}-26}{9} > 1.$$

It follows that

$$F_r(p) \geq F_r(P_r+1) \geq F_{33}(P_{33}+1) > 1.$$

This proves (9.3).

For the remaining inequalities, we use the elementary bound

$$W(N) < 3N^{1/3}.$$

Indeed,

$$\frac{W(N)^3}{N} \le \prod_{\substack{\ell \mid N \\ \ell \text{ prime}}} \frac{8}{\ell} \le \frac{8^4}{2 \cdot 3 \cdot 5 \cdot 7} < 27,$$

since each factor corresponding to a prime $\ell \ge 11$ is less than 1.

Put

$$t = \frac{\sigma \varphi(N)}{p}, \qquad \delta = 1 - \sigma = \frac{1}{10000}.$$

As N is even, we have $0 < t < 1/2$. The inequality (9.3), already proved, gives

$$t > 8 \left(\frac{2 \log^2(N/2)}{N} \right)^{1/26} > 8p^{-1/26}.$$

On the other hand, (9.1) is equivalent to

$$t^{3/2} > \frac{32W(N)}{\delta \sqrt{p}} \left(\frac{p}{N} \right)^2 \sqrt{\frac{16-t}{1-t}}.$$

Using $W(N) < 3p^{1/3}$, $(p/N)^2 < 2$, and $t < 1/2$, we find

$$\frac{32W(N)}{\delta \sqrt{p}} \left(\frac{p}{N} \right)^2 \sqrt{\frac{16-t}{1-t}} < \frac{768\sqrt{2}}{\delta} p^{-1/6}.$$

Since

$$p^{17/156} > X^{17/156} > 534668 > \frac{48}{\delta},$$

the last expression is less than $16\sqrt{2}\, p^{-3/52}$, which in turn is less than $t^{3/2}$. This proves (9.1).

Finally, elementary simplification yields

$$\frac{p-3}{2} - \frac{\sigma N^2}{2p} = \frac{\delta p}{2} + \sigma - \frac{3}{2} - \frac{\sigma}{2p} > \frac{\delta p}{2} - 1 > \frac{p}{30000}.$$

The function $x^{1/6}/L(x)$ is increasing for $x \ge X$, and

$$\frac{X^{1/6}}{L(X)} > 2.25 \times 10^7 > 90000.$$

Therefore

$$W(N) \sqrt{p}\, L(p) < 3p^{5/6} L(p) < \frac{p}{30000},$$

which proves (9.2).

□

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